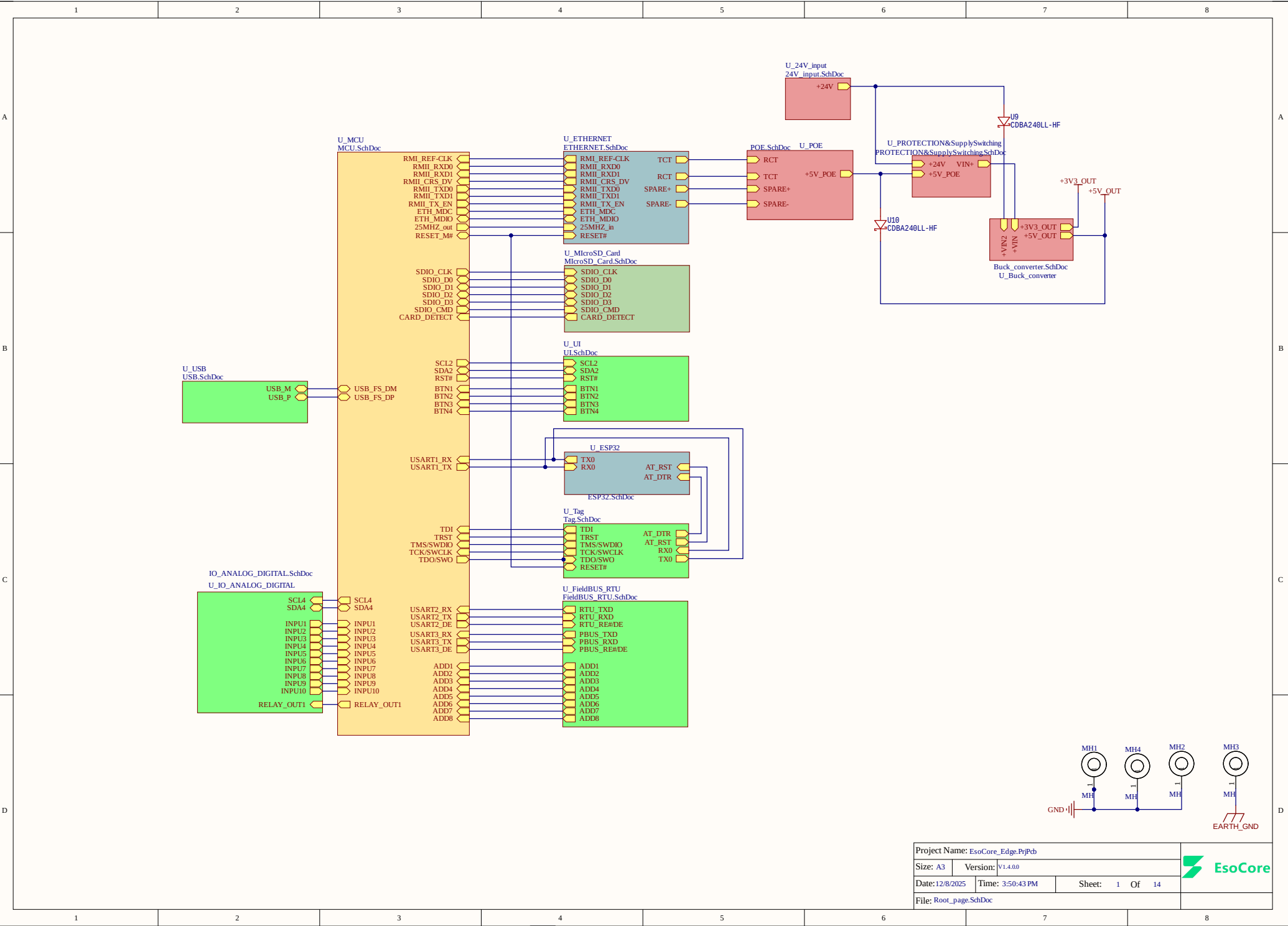
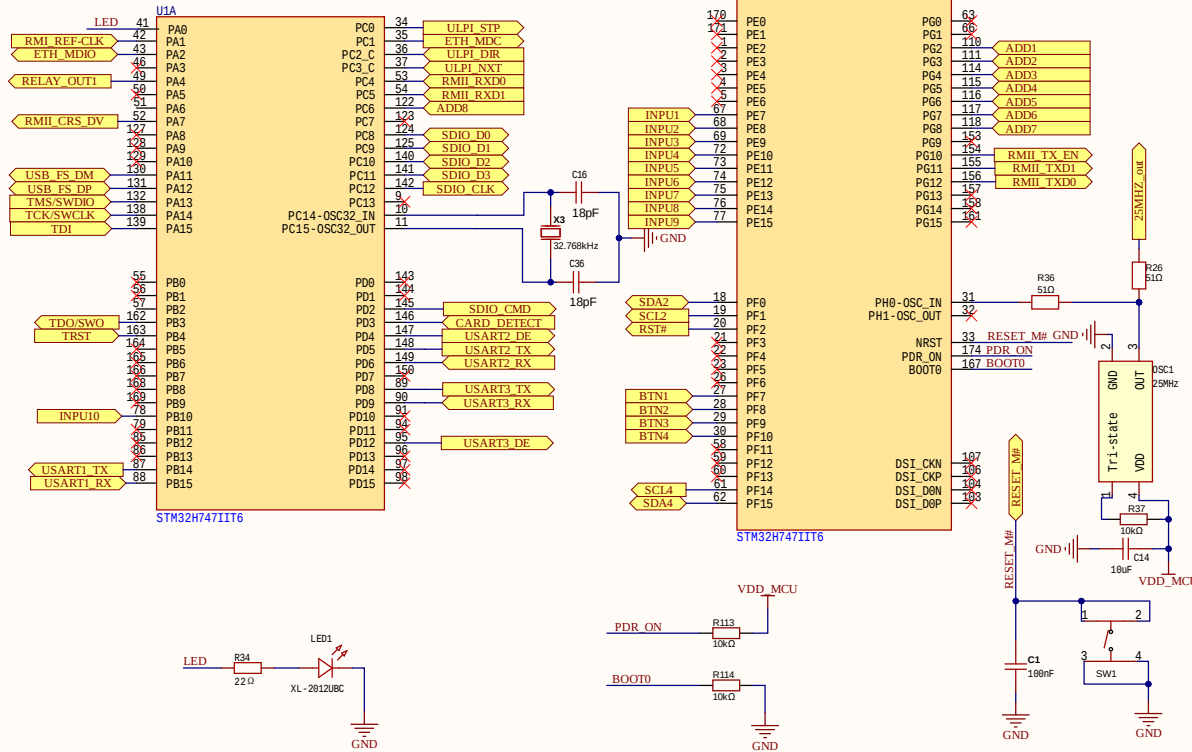


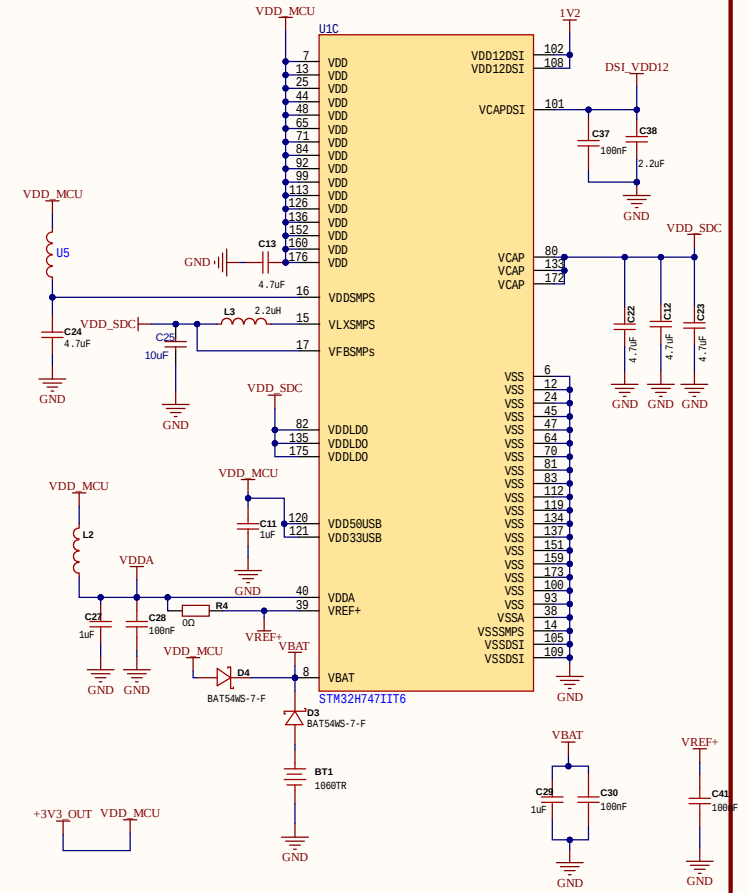


MCU

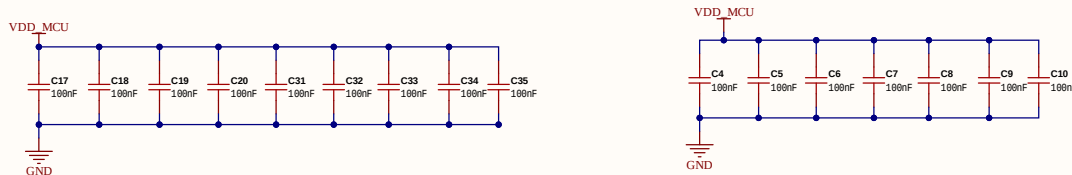
I/O PINS



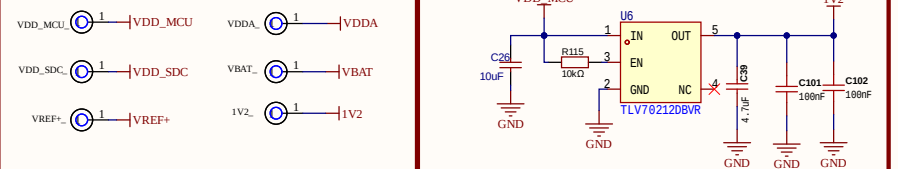
POWER PINS



POWER AND DECOUPLING CAPACITOR



Test point



Project Name: EsoCore_Edge_PjPcb

Size: A3 Version: V1.4.00

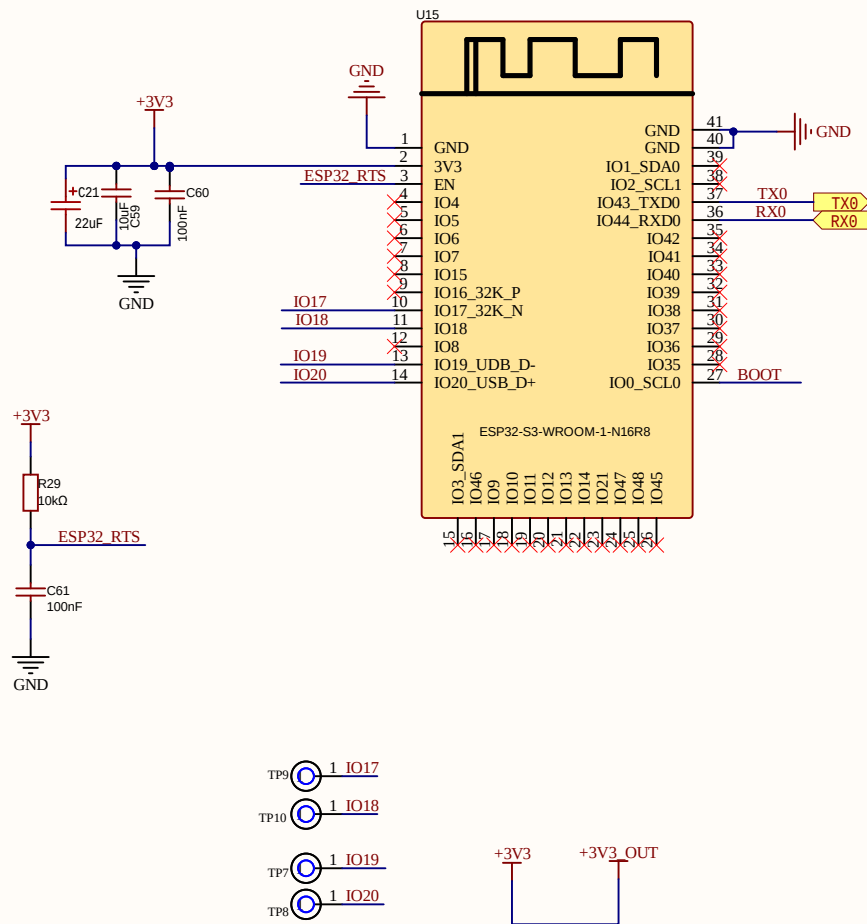
Date:12/8/2025 Time: 3:50:43 PM

File: MCU.SchDoc

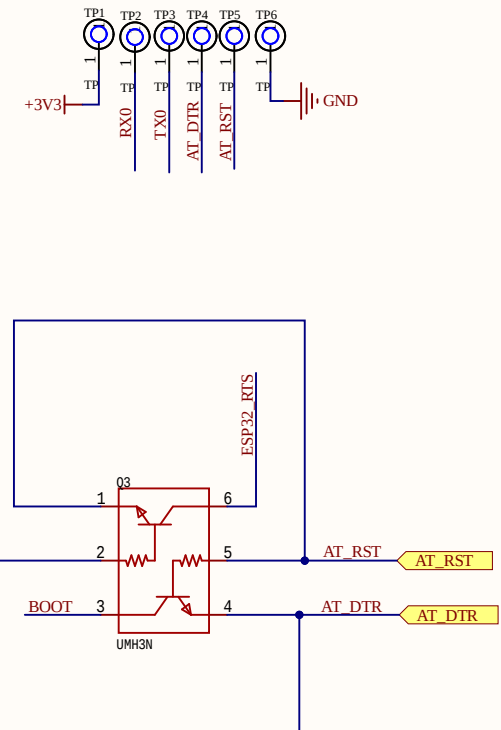
Sheet: 2 Of 14



ESP32



UART programmer(optional)



Project Name: EsoCore_Edge.PrjPcb

Size: A4 Version: V1.4.0.0

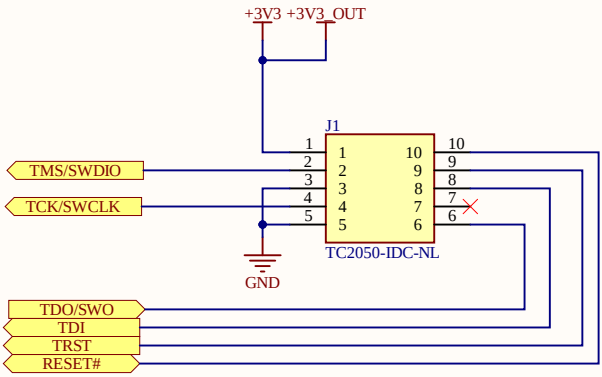
Date: 12/8/2025 Time: 3:50:43 PM

Sheet: 3 Of 14

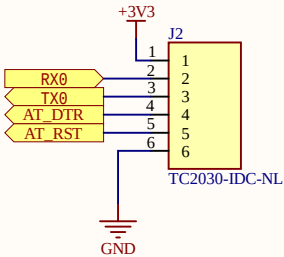
File: ESP32.SchDoc



STM32 TAG



ESP32 TAG



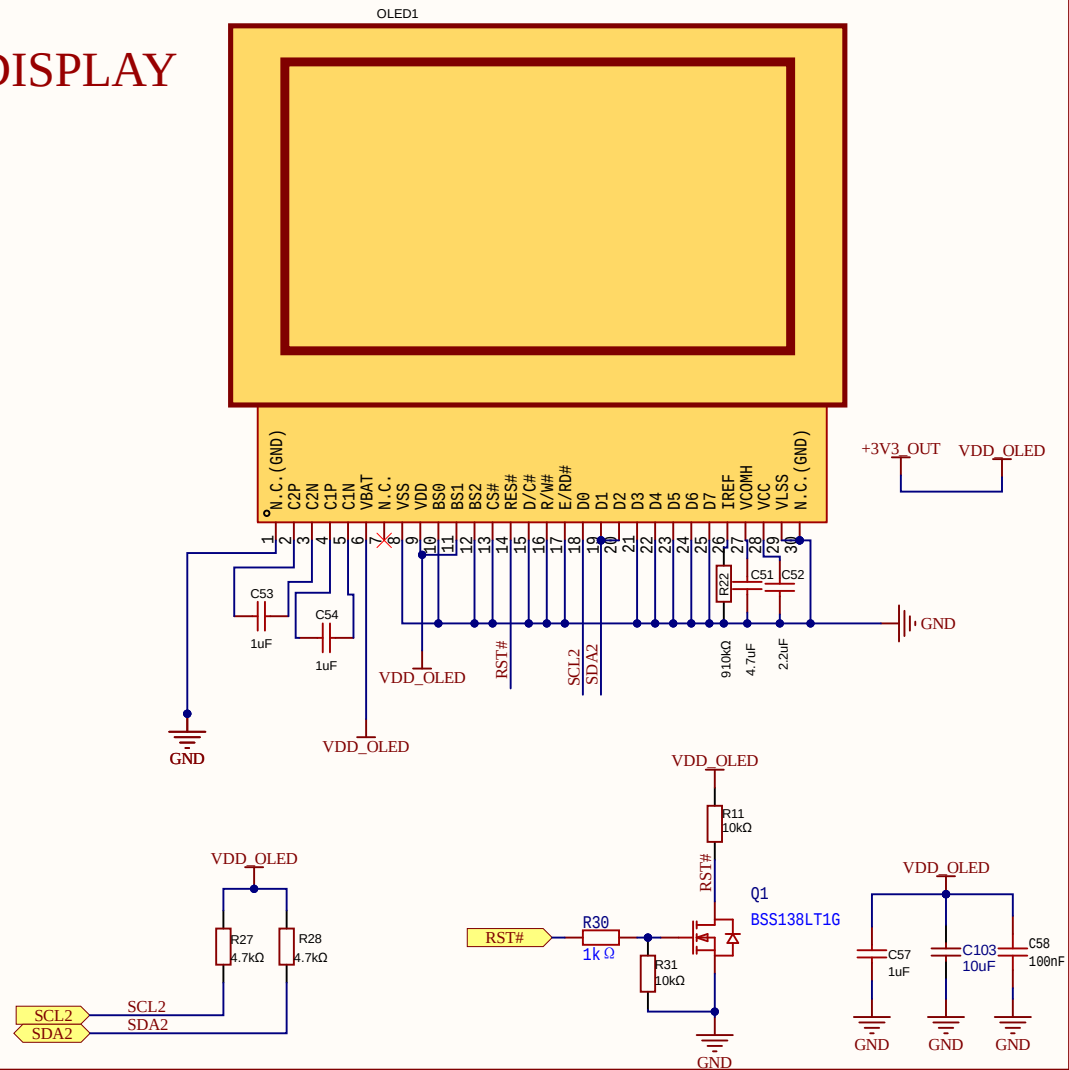
Project Name: EsoCore_Edge.PrjPcb					
Size: A4	Version: V1.4.0.0				
Date: 12/8/2025	Time: 3:50:43 PM	Sheet: 4 Of 14			
File: Tag.SchDoc					

The schematic diagram illustrates the SDIO interface circuit. It features two integrated circuits: the EMIF06-MSD02N16 (U7) and the DM3AT-SF-PEJM5 (CARD1). The EMIF06-MSD02N16 (U7) is connected to the SDIO signals (SDIO_D2, SDIO_D3, SDIO_CMD, SDIO_CLK, SDIO_D0, SDIO_D1) and the 3V_SD_card supply. The DM3AT-SF-PEJM5 (CARD1) is connected to the SD card interface signals (DAT2, CD/DAT3, CMD, VDD, CLK, VSS, DAT0, DAT1) and the 3V_SD_card supply. A CARD_DETECT signal is connected to the SW_A pin of CARD1. A decoupling circuit with capacitors C62 (100nF) and C63 (2.2uF) is connected to the 3V_SD_card supply. The +3V3_OUT supply is also shown.

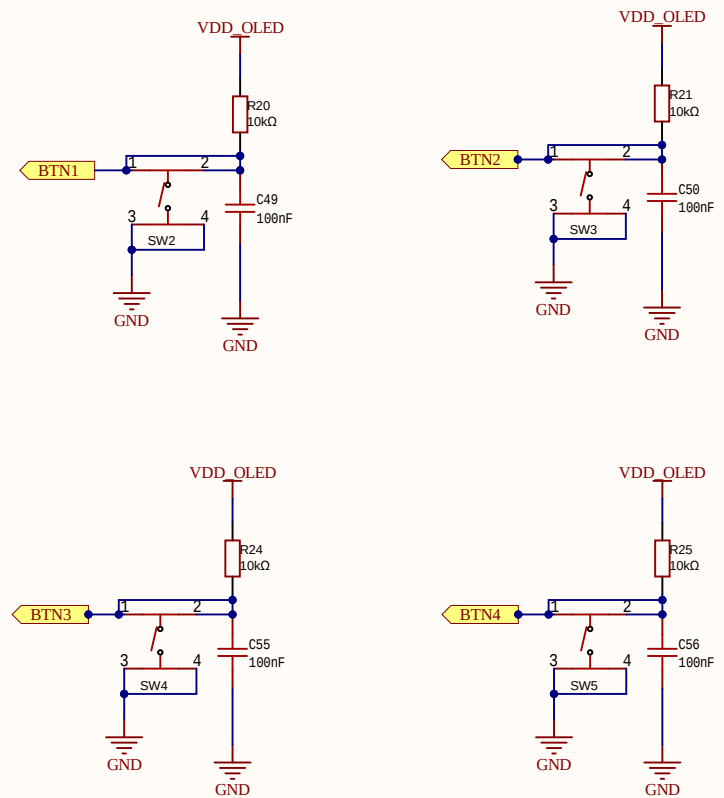
Project Name: EsoCore_Edge.PrjPcb			
Size: A4	Version: V1.4.0.0		
Date: 12/8/2025	Time: 3:50:44 PM	Sheet: 5	Of 14
File: MicroSD_Card.SchDoc			



DISPLAY



BUTTONS



Project Name: EsoCore_Edge.PrjPcb

Size: A4

Version: V1.4.0.0

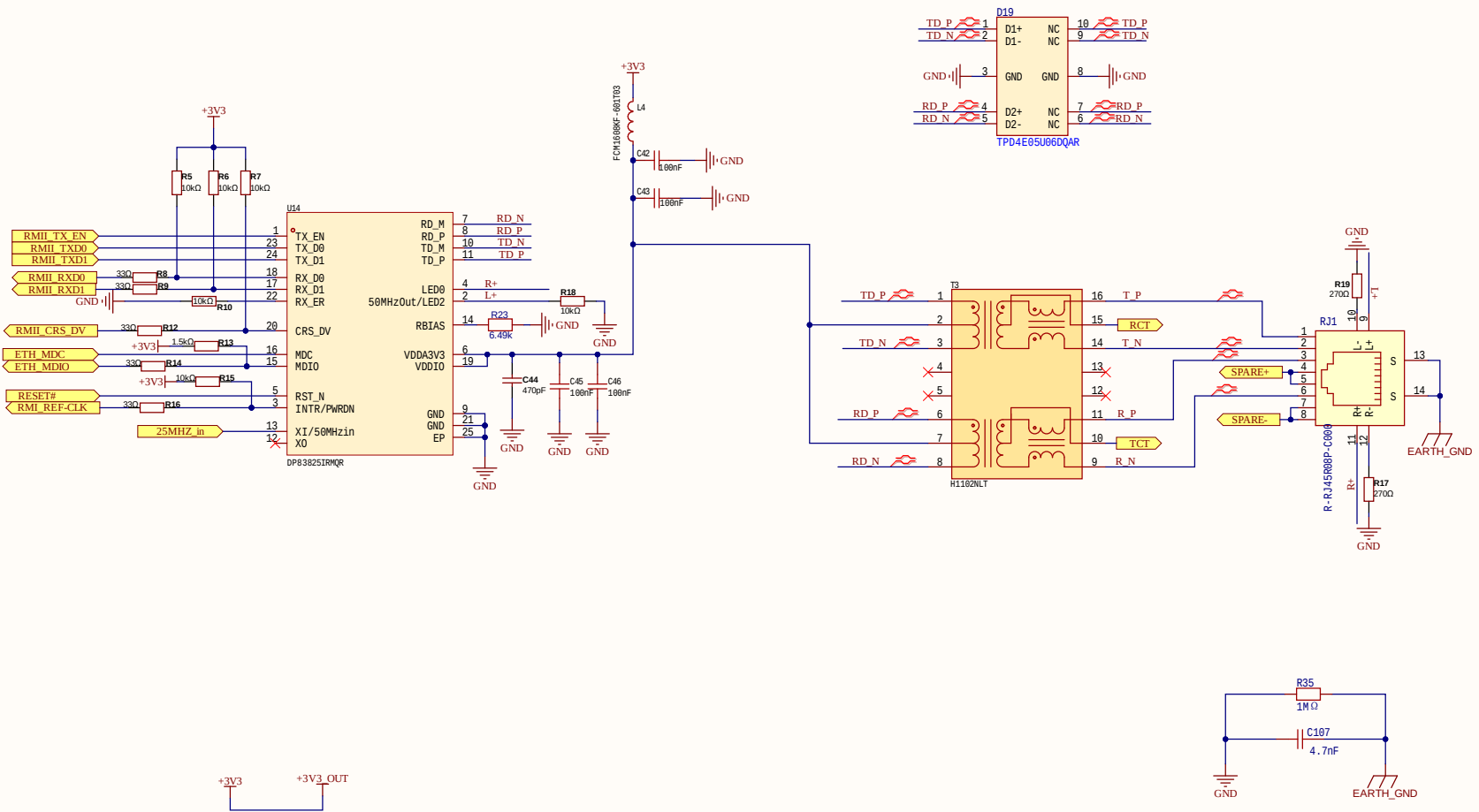
Date: 12/8/2025

Time: 3:50:44 PM

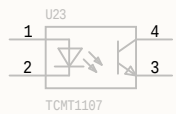
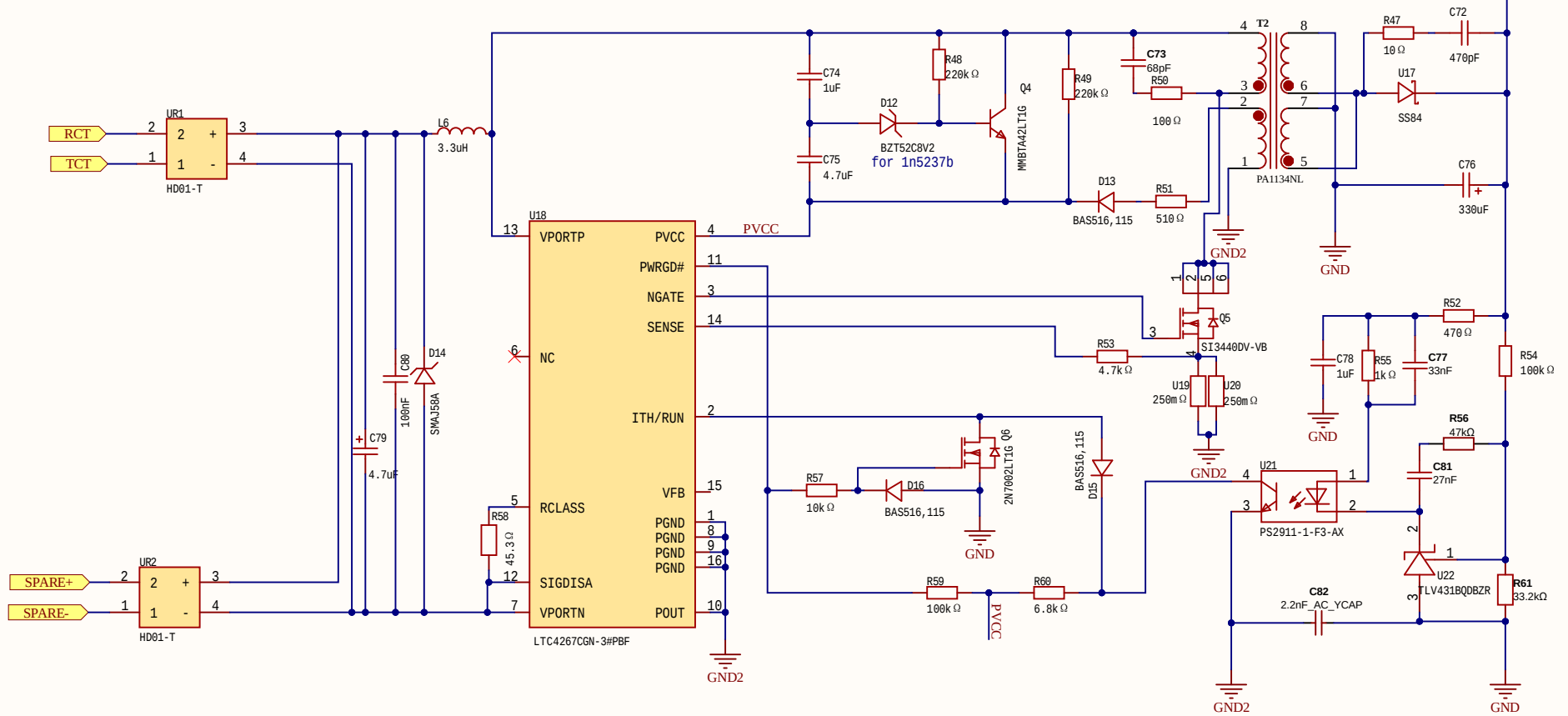
Sheet: 6 Of 14

File: UI.SchDoc

ETHERNET



POE



Project Name: EsoCore_Edge.PrjPcb

Size: A4 Version: V1.4.0.0

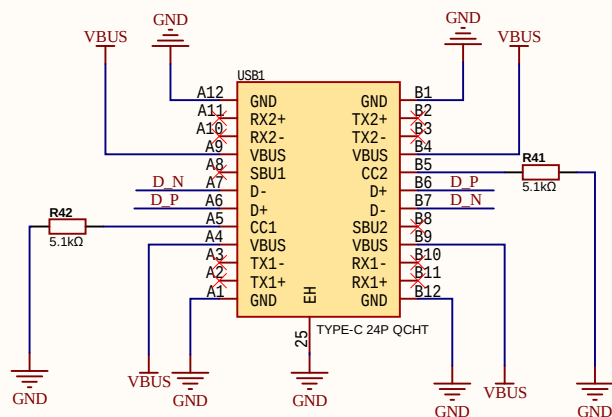
Date: 12/8/2025 Time: 3:50:44 PM

Sheet: 8 Of 14

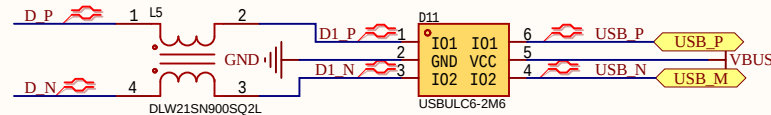
File: POE.SchDoc



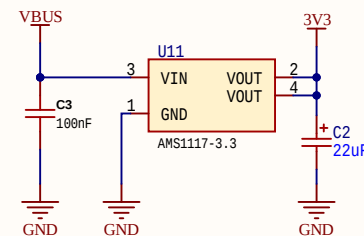
USB PORT



Common-Mode Choke & ESD



5V USB TO 3V3 SUPPLY



Project Name: **EsoCore_Edge.PrjPcb**

Size: **A4**

Version: **V1.4.0.0**

Date: **12/8/2025**

Time: **3:50:44 PM**

Sheet: **9** Of **14**

File: **USB.SchDoc**



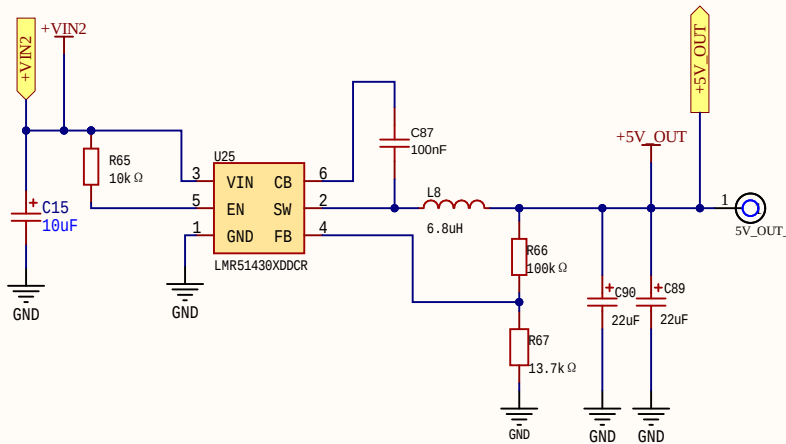
•



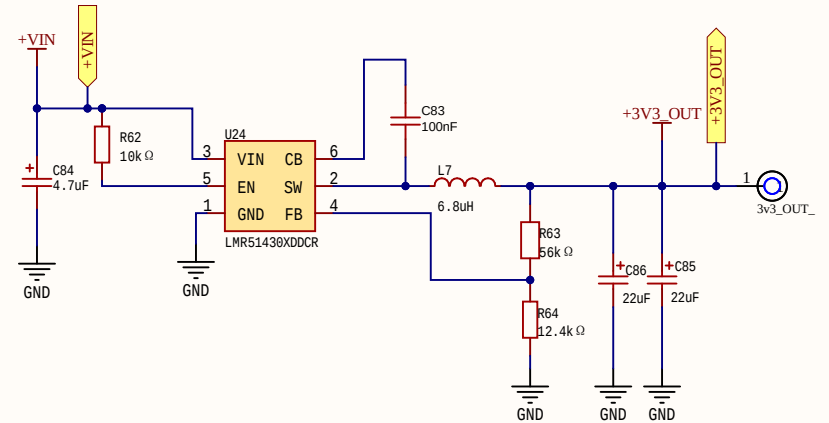
I



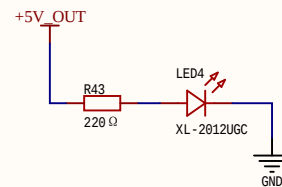
5V buck



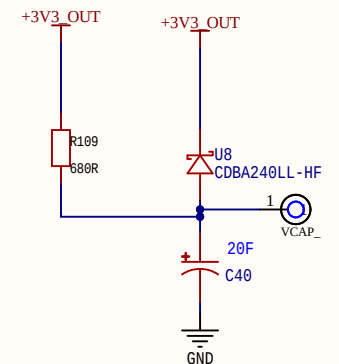
3v3 buck



Power LED



Super CAP



Project Name: **EsoCore_Edge.PrjPcb**

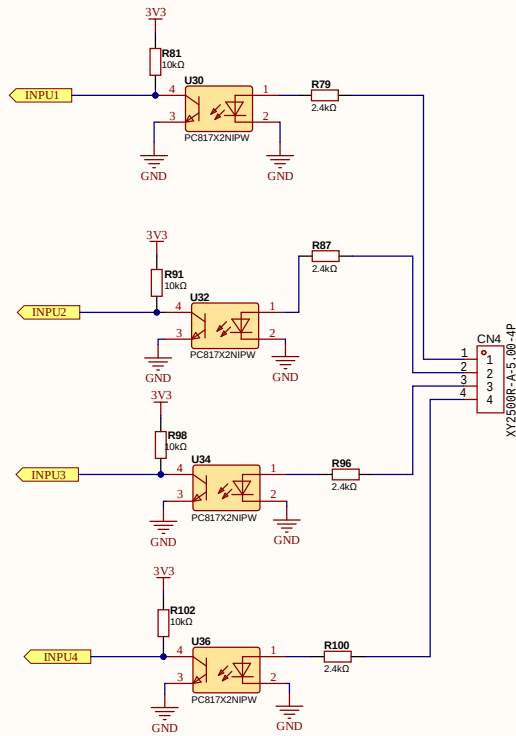
Size: **A4** Version: **V1.4.0.0**

Date: **12/8/2025** Time: **3:50:45 PM** Sheet: **11** Of **14**

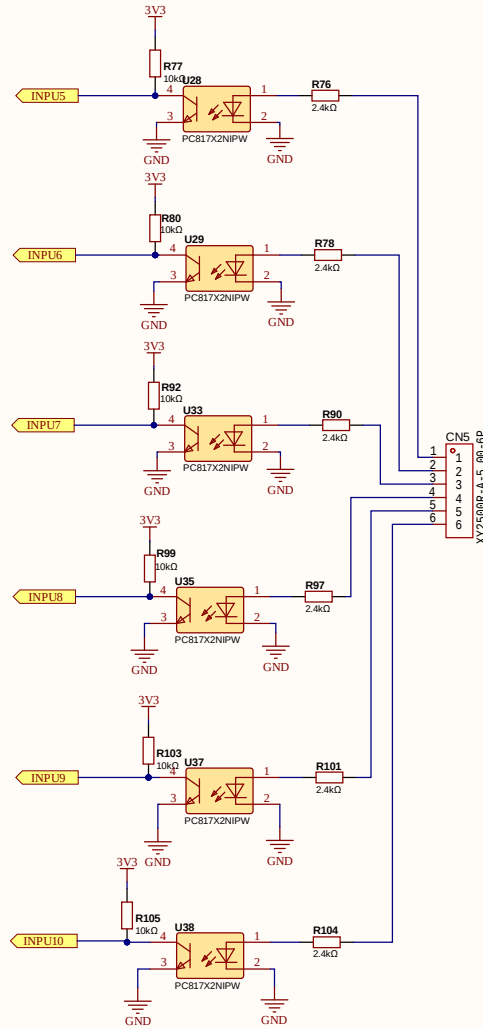
File: **Buck_converter.SchDoc**



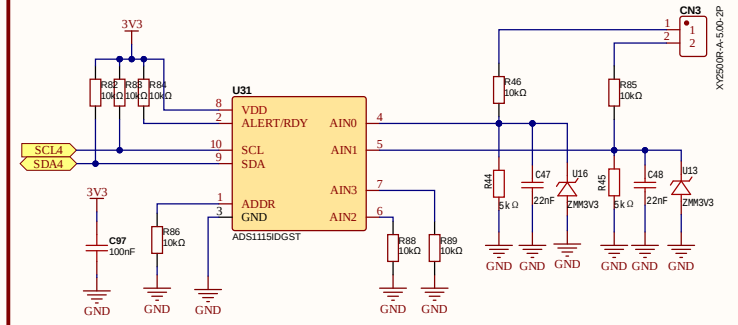
GENERAL INPUT



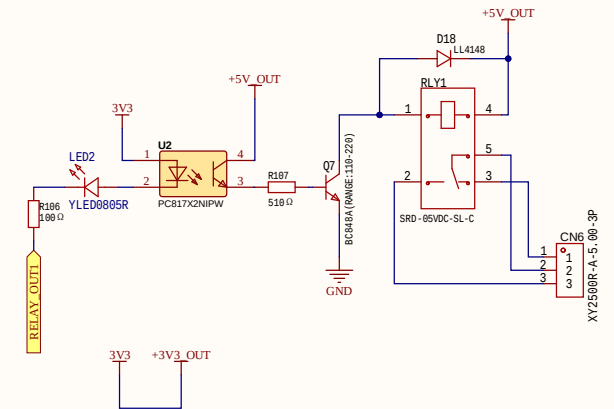
SAFETY INPUT



ANALOG INPUT



RELAY OUTPUT



CN6
XY2500R-A-5.00-8P

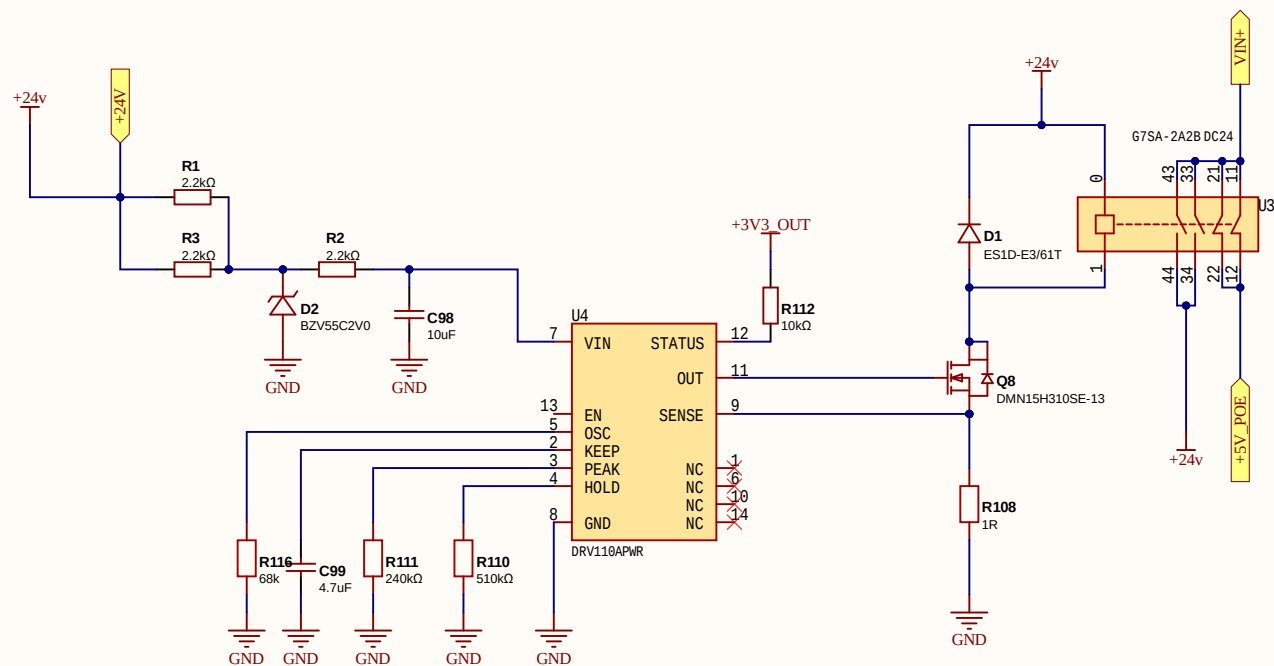
Project Name: EsoCore_Edge_PjPcb

Size: A3 Version: V1.4.0.0

Date: 12/8/2025 Time: 3:50:45 PM Sheet: 12 Of 14

File: IO_ANALOG_DIGITAL_SchDoc

SWITCHING SUPPLY



Project Name: **EsoCore_Edge.PrjPcb**

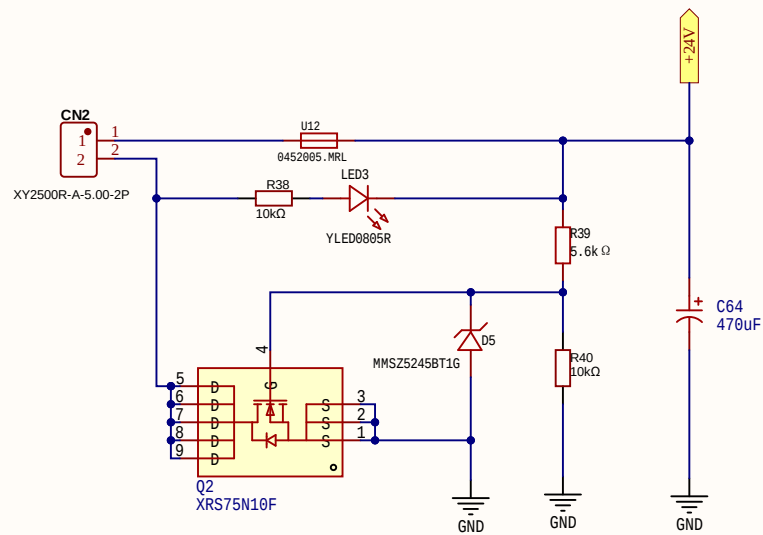
Size: **A4** Version: **V1.4.0.0**

Date: **12/8/2025** Time: **3:50:45 PM** Sheet: **13** Of **14**

File: **PROTECTION&SupplySwitching.SchDoc**



24V Input Terminal



Project Name: EsoCore_Edge.PrjPcb

Size: A4 Version: V1.4.0.0

Date: 12/8/2025 Time: 3:50:45 PM

Sheet: 14 Of 14

File: 24V_input.SchDoc

